



SECURITY INCIDENT INVESTIGATION REPORT

Linux Authentication & System Log Analysis



----- DETECT · ANALYZE · RESPOND · PROTECT -----

Prepared as part of a Security Operations Center (SOC) Incident Investigation project.

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 **PROJECT**

SOC Incident Investigation using Linux
Authentication and System Logs

 **PROJECT DOMAIN**

Security Operations Center (SOC)
Incident Response | Blue Team

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Objective

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The objective of this project is to investigate Linux authentication and system log files to identify suspicious activities, distinguish legitimate events from potential security incidents, develop detection logic, classify incidents based on severity, and recommend appropriate response actions following Security Operations Center (SOC) methodologies.

Project Overview

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This project simulates a real-world Security Operations Center (SOC) incident investigation using Linux authentication (auth.log) and system (syslog.log) log files. The investigation focuses on analyzing authentication events, user activities, scheduled tasks, and system logs to identify indicators of compromise (IOCs). Based on the collected evidence, suspicious activities are classified according to their severity, appropriate detection rules are developed, and incident response actions are recommended. The investigation follows the standard incident response lifecycle, including detection, investigation, containment, eradication, recovery, and closure, demonstrating the analytical approach used by SOC analysts when responding to security incidents.

Log Analysis

3.1 Authentication Log Analysis

The authentication log (auth.log) was analyzed to identify both legitimate system activities and potential security incidents. The log contains records of user authentication, SSH access, privilege escalation, session management, account creation, and password changes. During the initial analysis, normal administrative activities performed by legitimate users were identified alongside several suspicious events that indicate a possible system compromise.

The investigation began with normal user authentication activities involving the users `aritra` and `ubuntu`. These activities included successful SSH logins using public key authentication, creation of user sessions, execution of privileged commands through **`sudo`**, and normal session termination. Daily package update activities executed by the operating system were also observed and classified as legitimate administrative operations.

Further analysis identified a large number of failed authentication attempts targeting multiple user accounts. Between line 58 and 172 of the log, repeated login attempts were performed against the invalid user account `guest`, generating multiple “Invalid user” and “Failed password” events. Such repeated authentication failures indicate a possible brute-force attack attempting to discover valid usernames and passwords.

Following the failed attempts against the `guest` account, repeated failed login attempts were observed against the **`backup`** account. After several unsuccessful authentication attempts, the attacker successfully authenticated using valid credentials for the `backup` account. This successful login immediately following repeated failures strongly suggests a compromised account resulting from password guessing or credential compromise.

After obtaining access through the `backup` account, new user and group creation events were identified. A new user account named **`sysupdate`** was created, followed by password assignment and successful authentication using the newly created account. The creation of a new user account immediately after system compromise indicates an attempt to establish persistence, allowing continued unauthorized access even if the original compromised account is disabled.

Based on the authentication log analysis, multiple Indicators of Compromise (IOCs) were identified, including repeated failed login attempts, successful authentication following repeated failures, unauthorized user account creation, password assignment for the newly created account, and successful login using the persistence account. These events collectively indicate that the attacker successfully compromised the system and established persistent access.

3.2 System Log Analysis

The system log (syslog.log) was analyzed to identify system-level activities, scheduled tasks, service events, and any indicators of suspicious behavior. The analysis revealed both legitimate operating system processes and activities that may indicate unauthorized access or malicious persistence.

The initial log entries primarily consisted of normal system operations, including scheduled CRON jobs, service management activities, and automatic maintenance tasks. Regular execution of **logrotate** was observed, which is responsible for rotating and compressing system log files to prevent excessive disk usage. These events are expected administrative activities and classified as legitimate.

Further analysis identified periodic execution of **Certbot** renewal tasks. These scheduled jobs automatically renew SSL/TLS certificates before expiration to maintain secure HTTPS communication. Since certificate renewal is a standard Linux maintenance process, these events were also classified as normal system behavior. Routine service management activities performed by **systemd** were observed throughout the log. These included starting and stopping user sessions and managing background services required for normal system operation. No suspicious behavior was identified during these events.

However, later analysis identified a suspicious scheduled task executed through **CRON**. Unlike the legitimate maintenance jobs observed earlier, this task downloaded and executed a remote script from an external source. Executing code retrieved from an unknown external location is highly suspicious because it may allow an attacker to install malware, establish persistence, or execute unauthorized commands on the compromised system.

The investigation also identified outbound communication between the compromised system and an unknown external IP address. Such network connections may indicate command-and-control (C2) communication or data exfiltration and should be treated as a high-priority security event requiring immediate investigation.

Based on the system log analysis, the identified indicators strongly support the findings from the authentication log investigation. The malicious scheduled task and suspicious outbound communication suggest that the attacker not only gained unauthorized access but also attempted to maintain persistence and potentially communicate with external infrastructure after compromising the system.

4. Detection Logic

The following detection rules were developed based on the suspicious activities identified during the analysis of the authentication and system logs. These rules are intended to help a Security Information and Event Management (SIEM) solution automatically identify similar malicious activities in the future. By generating alerts for these events, security analysts can investigate potential threats more quickly and reduce the risk of successful system compromise.

4.1 Detection Logic

Rule ID	Detection Rule	Alert Condition	Reason for Alert
DR-001	Multiple Failed Login Attempts	More than 5 failed attempts for the same IP or against the same account within 5 minutes.	Indicates a possible brute-force or password guessing attack.
DR-002	Successful Login After Multiple Failures	A successful login occurs immediately after multiple failed authentication attempts	May indicate that an attacker successfully guessed or obtained valid credentials.
DR-003	New User Account Creation	A new user account is created on the system	Detects possible persistence mechanisms established by an attacker.
DR-004	Password Assignment for New User	A password is assigned immediately after a new amount is created	Indicates configuration of a newly created account for future unauthorized access.
DR-005	Suspicious CRON job Execution	CRON executes commands that download or execute files from external	May indicate malware installation or execution of unauthorized scripts.

		sources.	
DR-006	Outbound Connection to Unknown External IP	The system establishes communication with an unknown external IP address.	May indicate command-and-control communication or data exfiltration.

4.2 Explanation of Detection Rules

DR-001 - Multiple Failed Login Attempts

This rule monitors repeated authentication failures within a short period. Multiple failed login attempts targeting the same account or originating from the same IP address may indicate a brute-force or password guessing attack. Generating an alert allows the SOC team to investigate suspicious authentication activity before a successful compromise occurs.

DR-002 - Successful Login After Multiple Failed Attempts

This rule detects successful authentication immediately following several failed login attempts. Such behavior may indicate that an attacker successfully guessed valid credentials or used compromised passwords. This event requires immediate investigation because it suggests unauthorized system access.

DR-003 - New User Account Creation

This rule generates an alert whenever a new user account is created on the system. Attackers often create additional accounts after compromising a server to maintain persistent access. Monitoring user account creation helps identify unauthorized administrative activities.

DR-004 - Password Assignment for New User

This rule monitors password changes immediately following the creation of a new user account. Assigning a password to a newly created account may indicate that an attacker is preparing the

account for future use. Detecting this behavior enables security teams to verify whether the account creation was authorized.

DR-005 - Suspicious CRON Job Execution

This rule detects scheduled tasks that execute commands such as `wget` , `curl` , or other utilities to download and execute files from external sources. Since legitimate maintenance tasks rarely execute remote scripts, such behavior may indicate malware installation or persistence mechanisms and should be investigated immediately.

DR-006 - Outbound Connection to Unknown External IP

This rule monitors outbound network connections established with unknown external IP addresses. Unauthorized communication with external systems may indicate command-and-control (C2) communication, malware activity, or attempted data exfiltration. Immediate investigation is recommended to determine the purpose of the connection.

5. Incident Classification & Response

During the investigation, several suspicious activities were identified and classified according to their potential impact on the system. Each incident was assigned a severity level based on its likelihood of compromise, possible business impact, and the urgency of the required response. Appropriate response actions were recommended to assist the Security Operations Center (SOC) in containing and mitigating the identified threats.

5.1 Incident Classification

Incident ID	Suspicious Activity	Severity	Possible Impact	Recommended Response
INC-001	Multiple failed login attempts	Medium	Possible brute-force attack against user accounts	Monitor activity, block the source IP if the attempts continue, and

				investigate authentication logs.
INC-002	Successful login after multiple failed attempts (backup)	High	Possible account compromise and unauthorized access	Reset the account password immediately, terminate active sessions, and investigate user activity.
INC-003	Creation of new user account (sysupdate)	High	Establishment of persistence by an attacker	Verify whether the account creation was authorized. Disable or remove unauthorized accounts and notify the security team.
INC-004	Password assignment for the new account	High	Preparation of a persistence account for further access	Investigate the account creation process, remove unauthorized accounts, and review administrative activities.
INC-005	Suspicious CRON job downloading a remote script	High	Malware installation or execution of malicious code	Disable the scheduled task, isolate the affected system, and perform malware analysis.
INC-006	Outbound connection to an unknown external IP	High	Possible command-and-control communication or data exfiltration	Block the connection, isolate the host, investigate network traffic, and escalate the incident if malicious activity is confirmed.

INC-007	Automatic SSL certificate renewal (Certbot)	Low	Normal maintenance activity with no security impact	No response required. Continue routine monitoring.
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5.2 Incident Response Explanation

INC-001 - Multiple Failed Login Attempts

Repeated authentication failures were observed against multiple user accounts. Although no unauthorized access occurred during the initial attempts, such behavior is commonly associated with brute-force attacks. Continuous monitoring and blocking the offending IP address, if necessary, can help prevent successful compromise.

INC-002 - Successful Login After Multiple Failed Attempts

A successful login to the backup account occurred after several failed authentication attempts. This strongly suggests that valid credentials were obtained or guessed by an attacker. Immediate password reset, session termination, and investigation of account activity are recommended.

INC-003 - Unauthorized User Account Creation

A new user account named sysupdate was created shortly after the successful compromise of the backup account. Attackers commonly create additional accounts to maintain long-term access to compromised systems. Verification and removal of unauthorized accounts should be performed immediately.

INC-004 - Password Assignment for the New Account

Following the creation of the sysupdate account, a password was assigned, indicating that the account was prepared for future use. This activity further supports the presence of an unauthorized persistence mechanism and should be investigated immediately.

INC-005 - Suspicious CRON Job Execution

A scheduled CRON task executed a command that downloaded and ran a remote script from an external source. Such behavior is uncommon for legitimate maintenance tasks and may indicate malware installation or malicious automation. The affected system should be isolated while the task and downloaded files are investigated.

INC-006 - Outbound Connection to Unknown External IP

The compromised system established communication with an unknown external IP address. This behavior may indicate command-and-control communication or attempted data exfiltration. Immediate network investigation and containment actions are required to determine the purpose of the connection.

INC-007 - Automatic SSL Certificate Renewal

The system executed a scheduled Certbot renewal task to renew the SSL certificate before expiration. This is a standard Linux maintenance operation and does not indicate malicious activity. No response is required other than routine monitoring.

6. Incident Workflow

The incident response workflow describes the systematic approach followed by a Security Operations Center (SOC) to detect, investigate, contain, eradicate, recover from, and close a security incident. Following a structured incident response process helps minimize damage, reduce recovery time, and improve the organization's overall security posture.

6.1 Detection

The incident was initially detected during the analysis of the authentication (auth.log) and system (sys.log) logs. Multiple failed login attempts against several user accounts, followed by a successful login to the backup account, indicated a potential account compromise. Further investigation identified the creation of a new user account (sysupdate), password assignment, suspicious CRON job execution, and outbound communication with an unknown external IP address. These activities collectively suggested unauthorized access and possible persistence within the system.

6.2 Alert

Based on the detection logic, the Security Information and Event Management (SIEM) system would generate alerts for repeated failed login attempts, successful authentication following multiple failures, unauthorized user account creation, password assignment for newly created accounts, suspicious CRON job execution, and outbound connections to unknown external IP addresses. These alerts notify SOC analysts that immediate investigation is required.

6.3 Investigation

The SOC analyst reviews the authentication and system logs to determine the sequence of events. During the investigation, the analyst correlates evidence from both log files to identify how the attacker gained access, the actions performed after compromise, and any persistence mechanisms established on the system. The investigation confirms that the compromise involved unauthorized authentication, user account creation, and execution of malicious scheduled tasks.

6.4 Containment

To prevent further damage, the affected system should be isolated from the network. The compromised backup account should be disabled, and the unauthorized sysupdate account should be removed or disabled immediately. Malicious scheduled tasks should be stopped, and outbound communication with the suspicious external IP address should be blocked using firewall rules.

6.5 Eradication

After containment, the SOC team removes all malicious artifacts from the compromised system. Unauthorized user accounts are deleted, malicious CRON jobs are removed, downloaded scripts are deleted, and the system is scanned for malware or additional persistence mechanisms. Passwords for compromised accounts should be reset to eliminate unauthorized access.

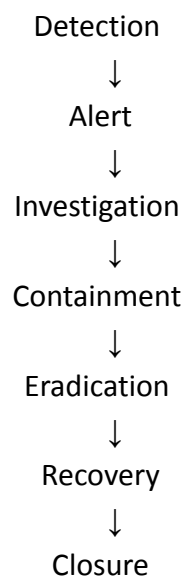
6.6 Recovery

Once the system has been verified as clean, normal services can be safely restored. User accounts should be monitored for suspicious authentication attempts, and security controls such as Multi-Factor Authentication (MFA), strong password policies, and continuous log monitoring should be implemented to reduce the likelihood of future compromises.

6.7 Closure

The incident is formally closed after confirming that all malicious activities have been removed, the affected systems are operating normally, and no further indicators of compromise are present. The investigation findings, response actions, lessons learned, and security recommendations are documented to improve future incident response capabilities and strengthen organizational security.

Workflow Diagram



7. Security Recommendation

Based on the findings of the investigation, several security improvements are recommended to reduce the risk of similar incidents in the future. These recommendations focus on strengthening authentication security, improving system monitoring, enhancing access control, and increasing the organization's overall incident detection and response capabilities.

7.1 Implement Multi-Factor Authentication (MFA)

Enable Multi-Factor Authentication (MFA) for all privileged and remote user accounts. Even if an attacker obtains valid credentials, MFA provides an additional layer of authentication that significantly reduces the likelihood of unauthorized access.

7.2 Enforce Strong Password Policies

Implement strong password policies requiring complex passwords, regular password changes, and account lockout mechanisms after multiple failed login attempts. These controls help mitigate brute-force and password guessing attacks.

7.3 Monitor Authentication Logs Continuously

Continuously monitor authentication logs to detect repeated failed login attempts, successful logins after multiple failures, and unusual authentication patterns. Automated alerting through a SIEM solution enables faster incident detection and response.

7.4 Restrict Administrative Privileges

Apply the principle of least privilege by granting administrative permissions only to authorized users. Regularly review privileged accounts to identify unauthorized account creation or privilege escalation.

7.5 Monitor User Account Creation

Configure security monitoring tools to generate alerts whenever new user accounts are created or existing accounts are modified. Unauthorized account creation is a common persistence technique used by attackers after compromising a system.

7.6 Audit Scheduled Tasks

Regularly review scheduled tasks and CRON Jobs to identify unauthorized commands or scripts. Unexpected downloads or execution of remote scripts should be investigated immediately as they may indicate malware installation or persistence mechanisms.

7.7 Implement Centralized Log Monitoring

Deploy a Security Information and Event Management (SIEM) solution such as Splunk, Microsoft Sentinel, Elastic Security, or Wazuh to centralize log collection, correlate security events, and automate threat detection through predefined detection rules.

7.8 Conduct Regular Security Audits

Perform regular security audits, vulnerability assessments, and log reviews to identify weaknesses before they can be exploited. Continuous security assessments improve the organization's ability to detect and respond to emerging threats.

8. Conclusion

This investigation successfully analyzed the provided Linux authentication and system log files to identify both normal system activities and potential security incidents. The analysis revealed evidence of repeated failed login attempts, successful authentication following multiple failures, unauthorized user accounts creation, suspicious scheduled task execution, and outbound communication with an unknown external IP address. Based on these findings, appropriate detection rules, incident classifications, incident response actions, and security recommendations were developed. The investigation demonstrates the importance of continuous log monitoring, timely incident response, and proactive security controls in protecting organizational systems against potential cyber threats.

9. Executive Summary

This report presents the findings of a Security Incident Investigation conducted using Linux authentication (auth.log) and system (syslog.log) log files. The objective of the investigation was to identify suspicious activities, classify security incidents, develop detection logic, and recommend appropriate incident response actions based on Security Operations Center (SOC) methodologies. The analysis identified multiple indicators of compromise, including repeated failed login attempts, successful authentication after multiple failures, unauthorized user account creation, malicious scheduled task execution, and suspicious outbound network communication. These findings indicate that the system was likely compromised and that persistence mechanisms were established. The report also provides detection rules, incident response procedures, and security recommendations to strengthen the organization's ability to detect, respond to, and prevent similar security incidents in the future.

10. References

Use these references:

1. MITRE ATT&CK Framework <https://attack.mitre.org/>
2. NIST Special Publication 800-61 Revision 2 - Computer Security Incident Handling Guide
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